ASSIUT - PHARMACY

بسم الله الرحمن الرحيم



Diarrhea in pediatrics



BY

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Definition and Epidemiology

➢ Definition:

Increase in fluidity, volume, and frequency of stool. Greenish stool without change in consistency is not diarrhea.

➢ Epidemiology:

The age incidence is between 3 months and 5 years. Diarrhea is common in summer and autumn, but less in winter and spring.



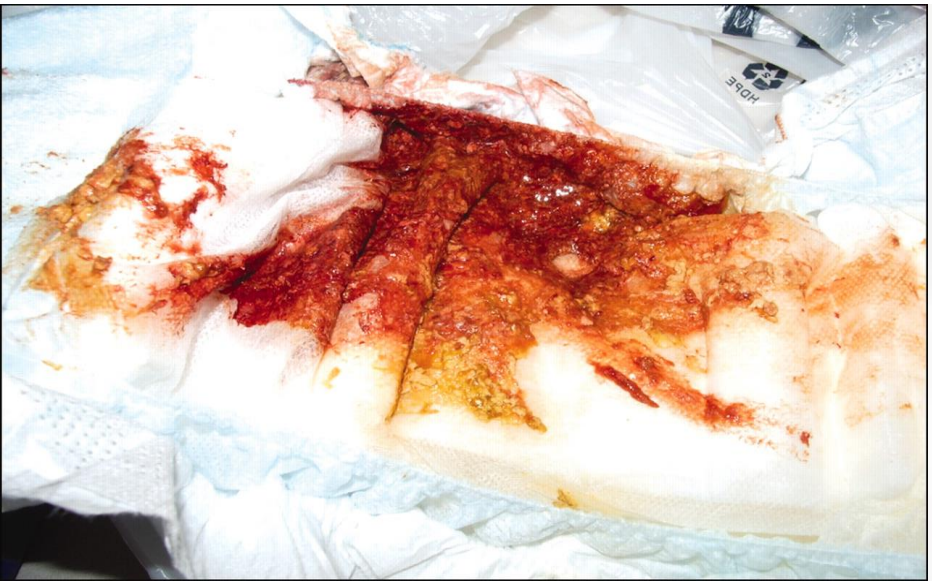


Clinical types

Clinical types

- ✓ Acute diarrhea: lasts less than 14 days
- ✓ Chronic diarrhea: lasts more than 14 days
- ✓ Dysentry: diarrhea + visible blood in stool

N.B. excessive mucus alone is not suggestive of dysentery.





Etiology

Causes of acute diarrhea:

The causes of acute diarrhea may be infectious or non infectious

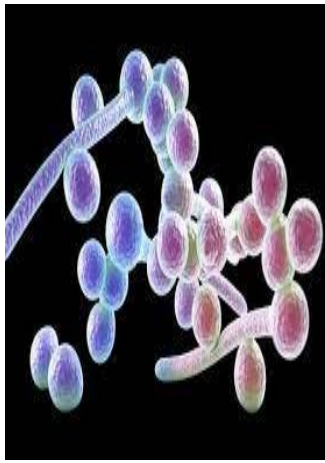
I- Infectious diarrhea

1- Bacterial: E-coli , Shigella, Cholera, and Salmonella.

2- Viral: Rotavirus, enterovirus.

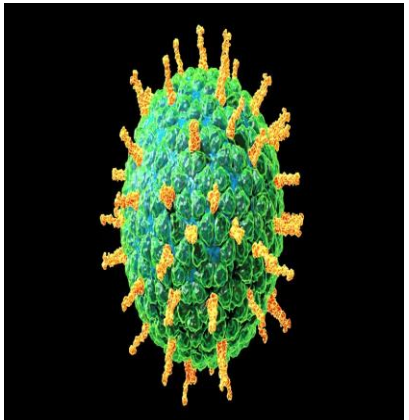
3- Parasitic: Entamoeba histolytica and Giardia lamblia.

4- Fungal: Candida.











Etiology

II- Non infectious diarrhea



1- Feeding problems e.g.:

- Overfeeding.
- Over concentrated formula in formula - fed infants.



2- Cow milk allergy.

3- Poisoning e.g. insecticides Poisoning.

4- Lactose intolerance.

Etiology

Causes of chronic diarrhea:

1- Infection:

A- Bacterial e.g. TB and Salmonella.

B- Parasitic e.g. E.h. and Giardia.

C- Viral e.g. Rota virus in immunocompromised patients.

D- Fungal: Candida.

2- Intestinal enzyme deficiency.

3- Cow milk allergy.

4- Prolonged use of antibiotics.

5- Endocrinal disorders: e.g. thyrotoxicosis.



Etiology

Causes of bloody diarrhea

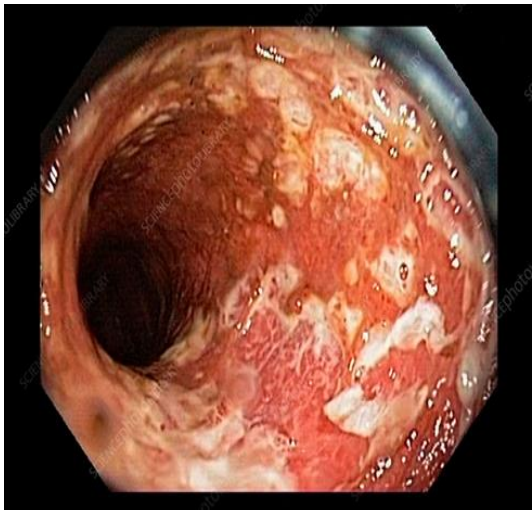
1- Infections e.g. shigella, bilharziasis, E. histolytica.

2- Cow milk allergy.

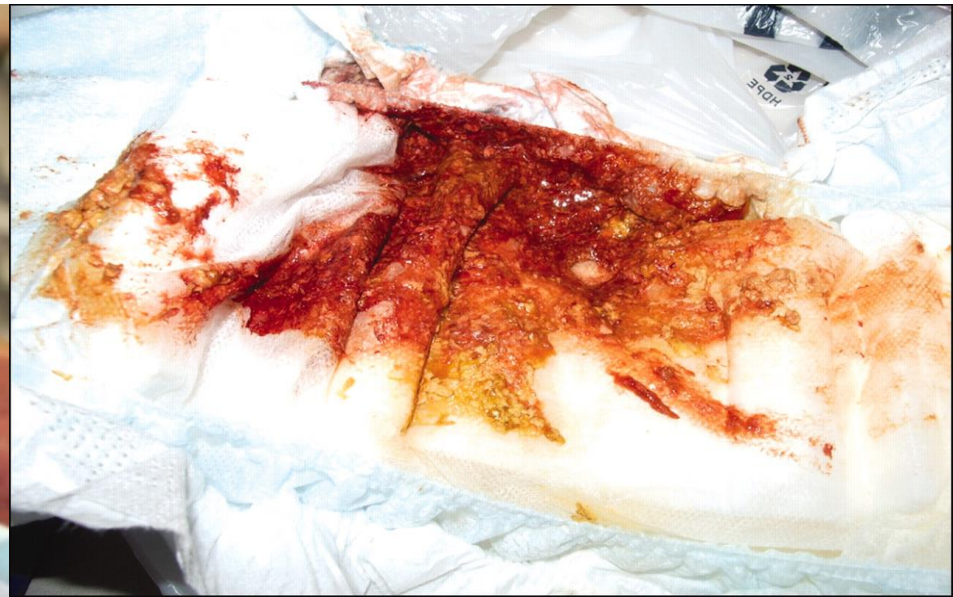
3- Bleeding tendency.

4- Intussusception.

5- Inflammatory bowel disease e.g. ulcerative colitis.







Risk Factors

I – Host Factors:

- Age
- Inadquate breast feeding
- Malnutrition
- Measles
- Vitamin A def.
- CMPA
- Antibiotic use
- Immunosuppression

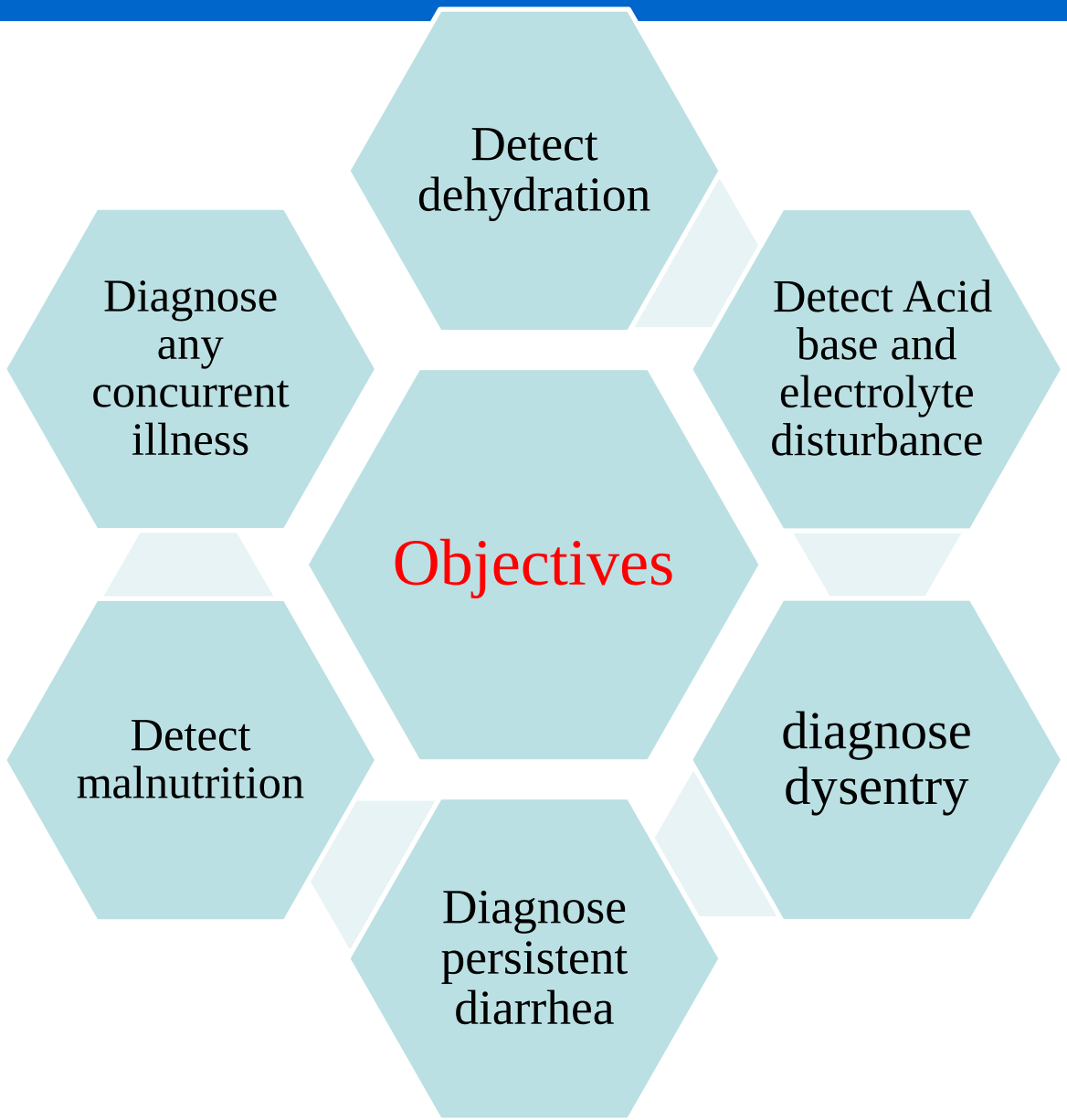




II – Behavioral Factors

- Contaminated formula or water
- Improper sewage disposal and poor sanitation

Clinical assessment



Complications

I- Dehydration:

Dehydration means deprivation of body fluids. It may lead to cardiovascular collapse and death.







Degrees and assessment of dehydration

Item	No dehydration Wt. loss<5%	Some dehydration Wt. loss 5 – 10 %	Sever dehydration Wt. loss >10%
General condition	Well, alert	Restless, irritable*	Lethargy, unconscious*
Eyes	Normal	Sunken	Very sunken, dry cornea
Tears	Present	Absent	Absent
Mouth and tongue	Moist	Dry	Very dry
Thirst	Drinks normal	Drinks eagerly*	Poor or unable to drink*
Skin pinch	Goes back quickly	Goes back slowly*	Goes back very slowly*
Treatment plan	Plan A	Plan B	Plan C

Other signs: pulse, blood pressure, skin, anterior fontanel, urine output

Complications

II- Metabolic and electrolyte disturbances:

include metabolic acidosis, metabolic alkalosis, hyponatremia, hypernatremia, hypokalemia, hypocalcemia, and hypomagnesemia.

III- Malnutrition:

Diarrhea produces malnutrition through:

- Decreased food intake due to anorexia
- Loss of nutrients in stool



Complications

IV- Convulsions:

May be due to metabolic and electrolyte disturbances e.g. hypo or hypernatremia, hypocalcemia, hypomagnesemia, and alkalosis.



V- Abdominal distention:

It may be due to:

- Paralytic ileus as a result of hypokalemia
- Intussusception



Complications

VI- Brain edema: may be due to:



- Hyponatremia
- Overhydration
- Rapid intravenous infusion

VII- Disseminated intravascular coagulopathy:

It leads to consumption of coagulation factors and liability to bleeding.

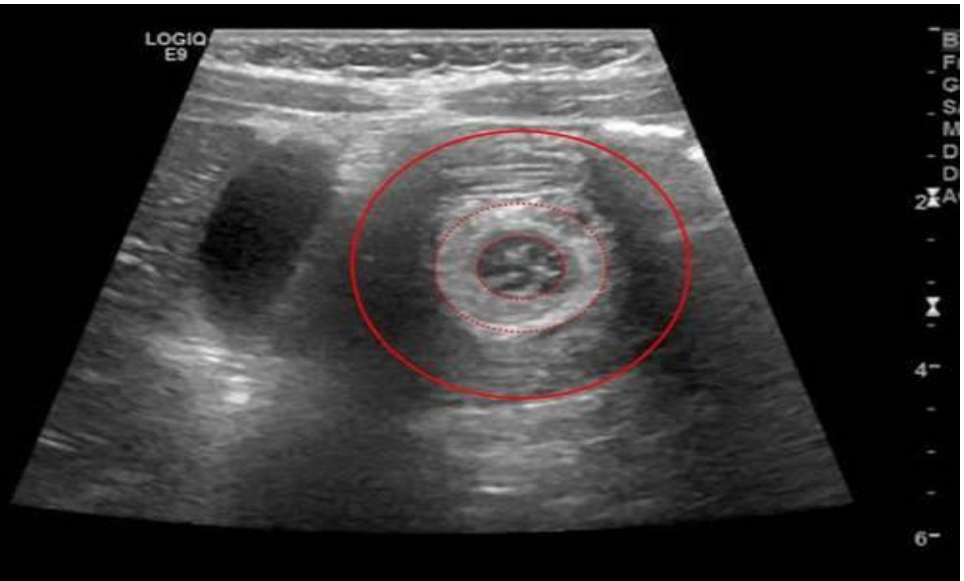
VIII- Acute renal failure: due to hypovolemia.

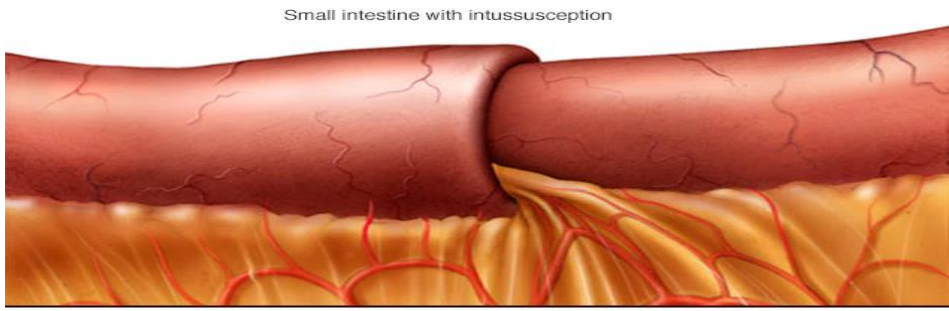
Complications

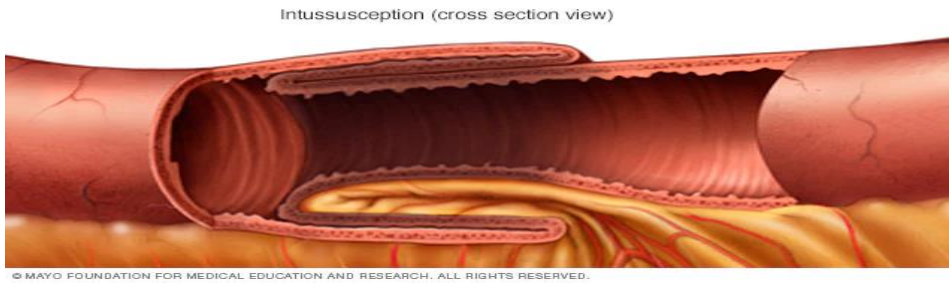
IX- Intussusception:

The child presents with abdominal distention, and colicky abdominal pain, preceding passage of red currant jelly stools. Intussusception may lead to intestinal perforation and peritonitis.

X- Septicemia







Treatment

Treatment of diarrhea includes:

A- Fluid therapy

B- Feeding

C- Additional therapies





Treatment

A- Fluid therapy

Types of fluids:

- Oral rehydration solution (ORS)
- Home-made fluids
- Intravenous fluids.







Types of fluids

- ORS

✓ Advantages

1- Effective in most cases

2- Suitable for all types of diarrhea and all types of dehydration

3- Used for all ages

4- Can be used even in the presence of vomiting.

5- Cheap and easily prepared.

✓ ORS is usually given as one teaspoon every minute.

Types of fluids

❑ Home made fluids:

These are fluids containing some salt with carbohydrates or protein, as soup and yoghurt. They should not be sweetened not to cause osmotic diarrhea.



❑ I.V. fluids:

- ➢ Polyvalent
- ➢ Ringer's lactate
- ➢ Normal saline 0.9%
- ➢ ½ normal saline 0.45%



Plans of fluid therapy

Plan A	Plan B	Plan C
No dehydration	Some dehydration	Severe dehydration
ORS or Home made fluids	ORS	I.V. fluids or N.G. rehydration
At home	At hospital	At hospital

Plan A

For cases with no dehydration.

The mother should be instructed to:

- ✓ Give the child more fluids (ORS or home made) to prevent dehydration.
- ✓ Give the child plenty of food (breast, usual formula, solid food, cereals, fruits) to prevent malnutrition.

Referral to hospital:

1- Becomes worse

2- Drinking poorly

3- Fever

4- Blood in stool





Plan B

For cases with some dehydration. ORS is given at hospital with assessment of progress of the patient.

1- Replacement of deficit in the first 4 hours

• The amount of ORS in the first 4 hours can be calculated as: weight in kg × 75 ml = …. ml.

• Breast feeding is encouraged, but no other food is offered during this period. If vomiting occurs, wait 10 minutes, then give ORS slowly every 3 minutes.

2- Assess after 4 hours:

• No signs of dehydration, shift to plan A.

• Signs of some dehydration, repeat plan B.

• Signs of severe dehydration, shift to plan C.

Plan C

- I.V. or N.G. rehydration
- Need hospitalization

❖ Indications for I.V. fluids:

1- Severe dehydration

2- Failure of oral or nasogastric rehydration

3- Persistent vomiting (>4 times /hour)

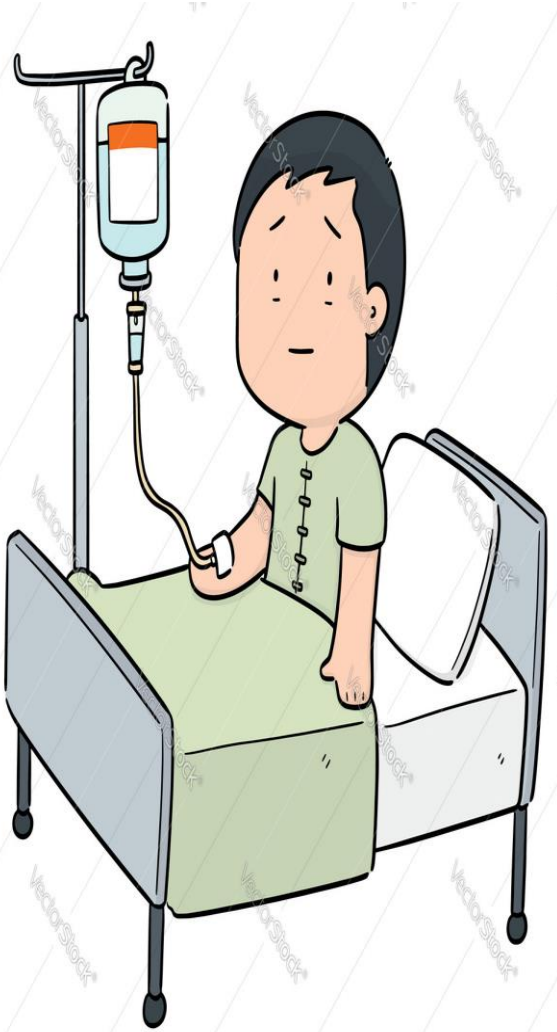
4- Convulsions

5- Severe abdominal distention



Plan C

- Small amounts of ORS are given by mouth as soon as the patient becomes able to drink.



Complications of IV rehydration:

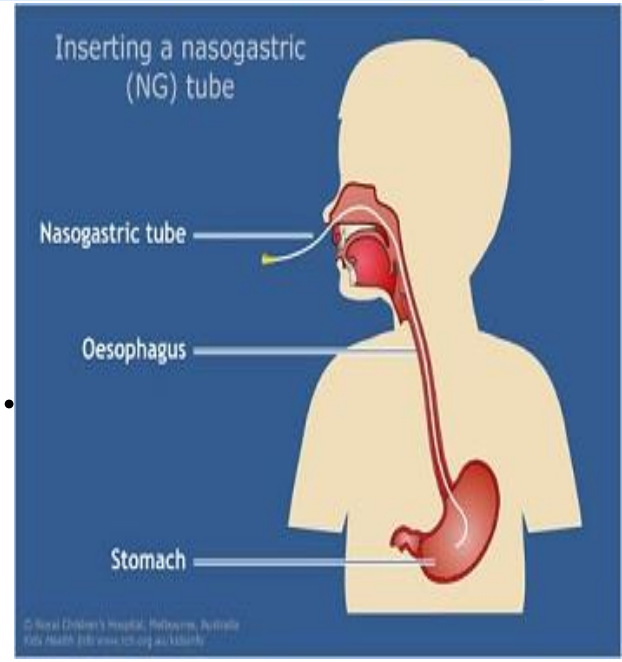
1- Electrolyte disturbances

2- Overhydration

3- Brain edema

4- Infection

Nasogastric rehydration



❑ Indications:

1. Failure of oral rehydration.
2. Persistent vomiting (>4 times /hour).
3. Severe anorexia or inability to swallow.
4. When the child refuses taste of ORS.
5. When the mother is tired.
6. In severe dehydration and I.V. fluids not available.

➢ Amount of fluid: 20 ml/kg/hr for 6 hr (total 120 ml/kg)

➢ Rate of drip = Wt. (kg) x 5 drop/minute

✓ The child should be reassessed every 2 hours

Feeding during diarrhea

- ✓ Early feeding and avoidance of fasting should be undertaken during treatment of diarrhea. Feeding during treatment of diarrhea does not increase the severity of diarrhea, but hastens recovery.
- ✓ Advantages of continued feeding:

1- Stimulate secretion of digestive enzymes.

2- Enhance salt and water absorption.

3- Enhance repair and function of intestinal mucosa.

4- Prevents malnutrition.

Feeding during diarrhea

- Types of feeding

1- Breast feeding:

2- Formula feeding: is continued in full concentration.

3- Solid and semisolid foods: should be small, frequent, easily digestible as; rice, yoghurt, and vegetable soups.







Additional therapies

1- Antimicrobial therapy: is indicated in:

a- Parenteral diarrhea which complicates bacterial infections outside GIT such as; acute otitis media, and pneumonia.

b- Specific enteric bacterial infections such as; shigella, salmonella, and cholera.



Additional therapies

2-Antipyretics: e.g. acetaminophen in case of fever, but the best antipyretics are good rehydration and cold compresses or water bath.



3- Anticonvulsants: in case of convulsions.

N.B. Antidiarrheal and antimotility drugs should not be given, as they may lead to paralytic ileus, and decrease elimination of microorganisms and their toxins.

Prevention of diarrhea



1- Promotion of breast feeding.

2- Proper water supply.

3- Proper sewage disposal.

4- Washing hands.

5- Measles vaccination.

6- Improvement of nutritional status.

7- Rota virus vaccination.


Thank You!